Learning Check

1. (True or False.) Every force that acts on anything has a specific direction associated with it.
2. Which of the following is *not* a unit of force?
   (a) newton (b) kilogram (c) ounce
   (d) ton (e) All are units of force.
3. The force of gravity acting on an object is called ______________.
4. The air exerts a force on a person standing in a strong wind. This type of force is one example of ______________ friction.

ANSWERS: 1. True 2. (b) 3. weight 4. kinetic

Learning Check

1. For an object to move with constant velocity,
   (a) the net force on it must be zero.
   (b) a constant force must act on it.
   (c) it must move in a circle.
   (d) there must be no friction acting on it.
2. (True or False.) It is possible for the net force on an object to be zero even if several forces are acting on it at the same time.
3. The force that must act on a body moving along a circular path is called the ______________.

ANSWERS: 1. (a) 2. True 3. centripetal force

Learning Check

1. If object A has a larger mass than object B,
   (a) A will be easier to accelerate than B.
   (b) A will weigh less than B.
   (c) A will be harder to keep moving in a circle.
   (d) All of the above.
2. (True or False.) The mass of an object depends on where the object is.

ANSWERS: 1. (c) 2. False

Learning Check

1. When a given net force acts on a body, the body's acceleration depends on its ______________.
2. The weight of an object equals its mass times ______________.
3. Identical cars A and B go around the same curve, with A traveling two times as fast as B. The centripetal force on A
   (a) is two times as large as the centripetal force on B.
   (b) is four times as large as the centripetal force on B.
   (c) is one-half as large as the centripetal force on B.
   (d) is the same as that on B, because they are on the same curve and have the same mass.

ANSWERS: 1. mass 2. acceleration of gravity 3. (b)

Learning Check

1. (True or False.) The speed of a projectile remains constant as it travels even though its velocity is changing.
2. The motion of a child on a swing moving back and forth is an example of ________.
3. A napkin is dropped and falls to the floor. As it falls,
   (a) the force of air resistance acting on it increases.
   (b) its speed increases.
   (c) the net force on it decreases.
   (d) All of the above.
4. In which of the following situations is the net force constant?
   (a) simple harmonic motion
   (b) freely falling body
   (c) a dropped object affected by air resistance
   (d) All of the above.

ANSWERS: 1. False **2.** simple harmonic motion **3.** (d) **4.** (b)

Learning Check

1. (True or False.) At this moment you are exerting a force upward on Earth.
2. In the process of throwing a ball forward
   (a) the ball exerts a rearward force on your hand.
   (b) the ball exerts a forward force on your hand.
   (c) your hand exerts a rearward force on the ball.
   (d) the only force present is your force on the ball.
3. At one moment in a football game, player A exerts a force of 200 N to the east on player B. The size of the force that B exerts on A is ________, and it is directed to the ________.
4. (True or False.) When you jump upward, the floor exerts a force on you that makes you accelerate.

ANSWERS: 1. True **2.** (a) **3.** 200 N, west **4.** True

Learning Check

1. (True or False.) If the distance between a spacecraft and Earth is doubled, the gravitational force on the spacecraft will be one-half as large.
2. (Choose the *incorrect* statement.) The gravitational force exerted on a satellite by Earth
   (a) is directed toward Earth's center.
   (b) depends on the satellite's speed.
   (c) depends on the satellite's mass.
   (d) depends on Earth's mass.
3. (True or False.) Gravity supplies the centripetal force needed to keep a planet in a circular orbit.
4. Every object creates a ________ in the space around it.

ANSWERS: 1. False **2.** (b) **3.** True **4.** gravitational field